

Multidimensional Analysis of Social Discourse on Twitter around Venezuelan Migration in Colombia

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Introduction

The outflow of refugees and migrants from Venezuela is one of the largest displacement crises in the world, with almost 7.7 million migrants and refugees as of November 2023 from which 6.5 million (84%) are currently in Latin America and the Caribbean, and 2.88 million (37%) in Colombia, making it the country that received the highest inflow [1]. Despite having a shared language, religion, and broad cultural heritage, this mass migration across Latin America has coincided with a decrease in migrant sentiment [2], and the formation of xenophobic, sexist, and discriminatory stances of the locals against the Venezuelan migrants [3]. These stances are influenced by ambivalent and even contradictory opinions and information from social networks, triggering xenophobic behaviors and actions [3].

Understanding how locals perceive migrants is crucial for several reasons. First, it can guide the development of inclusive policies by incorporating public opinions and concerns, which can influence immigration laws, social integration programs, and support services. Second, analyzing sentiment can reveal areas where community relationships are strained, highlighting where interventions are needed to promote migrant integration. Third, it helps counter misinformation and stereotypes about migrants, fostering greater social cohesion. Our study will examine this conversation across the dimensions of time, location, social interactions, and the use of xenophobic language.

Merging the analysis of past research through different lenses, our study seeks to analyze the discourse surrounding Venezuelan migration by examining it across three key dimensions: time, space [4], and the use of xenophobic terms [5]. By studying the conversation over time, we aim to understand how it has evolved, particularly in response to significant events, policies, and trends. Although the roots of this phenomenon date back to before 2015, the dramatic increase in migration since that year has heightened its relevance. Events such as the COVID-19 pandemic, the 2018 Presidential Elections, and the introduction of policies like the ETPV have likely influenced the discourse. Geographically, we aim to explore whether the location of individuals or the regions in which they reside, impacts the nature of the conversation. Lastly, we will investigate the use of xenophobic terms, which, although historically neutral, have taken on discriminatory connotations. By analyzing the frequency and tone of these terms, we aim to uncover patterns in their usage and the sentiments they convey over time.

Background and Literature

Venezuelan Migration

Other studies have examined the perception of locals about Venezuelan migrants in Colombia. For example, the Barómetro de Xenophobia [6] analyzed the conversation with xenophobic content on Twitter, but this was only done in 2021 and for the five cities of Bogotá, Cali, Medellín, Barranquilla, and Cúcuta. This study provided a limited snapshot of the discourse, as it focused on a small geographic area and a single year.

In contrast, the current study aims to provide a more comprehensive and multidimensional analysis of the social discourse surrounding Venezuelan migration in Colombia. By examining the conversation across a broader range of dimensions, including time, location, and the use of xenophobic language, the researchers hope to gain a deeper understanding of how the perception of Venezuelan migrants has evolved and how it varies across different regions

of the country. This approach allows for a more nuanced and holistic examination of the complex social dynamics at play, going beyond the limitations of previous studies that have focused on a narrower scope.

Use of Twitter and NLP for this analysis

Previous studies on the perception of Venezuelan migrants in Colombia have the limitation of not providing a comprehensive, multidimensional analysis of the social discourse. They have tended to offer a snapshot of the conversation, rather than examining how it has evolved in response to significant events, policies, and trends. Furthermore, the use of geolocation data has been limited, as many Twitter users do not have this feature activated. This has resulted in an incomplete picture of the geographic distribution of the discourse.

To address these limitations, the current study proposes a more robust and multifaceted approach. The researchers have compiled a much larger dataset of over 7.4 million tweets related to Venezuelan migration in Colombia, spanning a longer time period from 2014 to 2022. By training specialized language models in Spanish, they have been able to analyze the content of tweets regardless of whether geolocation data was available. This allows for a more comprehensive understanding of the discourse, not just in the major cities, but across the entire country.

Additionally, the researchers have adopted a multilayered approach, examining the conversation through the lenses of time, space, and the use of xenophobic language. This multidimensional analysis provides a more nuanced and holistic perspective on how the perception of Venezuelan migrants has shifted and varied across different regions of Colombia. By considering these various factors, the study aims to uncover patterns and insights that may have been missed in previous, more limited investigations.

Pejorative terms

Words or colloquialisms describing groups of people are not generally created by them but can often be reclaimed later as a method of empowerment. When a group is rendered physically or socially powerless—as in the case of Venezuelan migrants forced to leave their home country—the dominant culture describes the people and their predicament in words and phrases that capture out-group stereotypes. One scholar explains of the destructive nature of cultural expressions, “As a medium of communication, language expresses hidden notions of power, although, at a superficial level, the ideas and meanings contained in ordinary words are often assumed to be universally accepted by those who speak the language” [7]. In this paper, we explore how historically normalized words for Venezuelan migrants in Colombia, namely *veneco* and *veneca*, have transformed through social media to become markers of hate speech and xenophobic rhetoric on Twitter.

One striking and worrying aspect of the study is the prevalence of sexism among the populations, which translates into certain stereotypes about migrant women. Close to half of the people consulted in the three countries think that migrant women will end up engaging in prostitution; at the same time, sexist roles are being reproduced, leaving women overburdened with care responsibilities, which increases the likelihood of their rights being infringed [3]. *Veneco* slang for a Venezuelan man is used as an insult and *veneca* has outright become a synonym for prostitute [8].

Since the 1970s, Venezuela has been a safe haven for Colombians fleeing widespread civil conflict in hopes of work opportunities and a better life [9]. This amicable relationship and relatively porous border allowed Colombians to traverse between border towns with little immigration enforcement. During the 1970s when Venezuelans hosted millions of Colombians, the words *veneco* (masculine) and *veneca* (feminine) “weren’t always taken as an offense by Venezuelans.” Sergio Chacón, Master in Linguistics and Spanish, explains that this word arose from a kind of cultural and linguistic syncretism born within the border between both countries” [10]. The term, in its inception described Colombians that lived in Venezuela and developed a new accent, whereas now, it is a descriptor for Venezuelans migrating to Colombia [3].

Mid-2015 marked a reversal in migrant patterns. Venezuelans, and Colombians who had been in Venezuela for decades, began migrating to Colombia. “An estimated 4.5 million Venezuelans have fled their nation’s economic and humanitarian catastrophe in recent years, according to the U.N. About half of those are now residing in just two countries: Colombia and Peru” [4]. Corruption and a failed economy, often scapegoated as a botched socialism experiment, have led to an economic and political collapse that thrust Venezuelan citizens into the middle of a

humanitarian crisis with little access to basic necessities to sustain life [11]. This has led to a major outflow of Venezuelans migrating into neighboring countries like Colombia (see Figure 1).

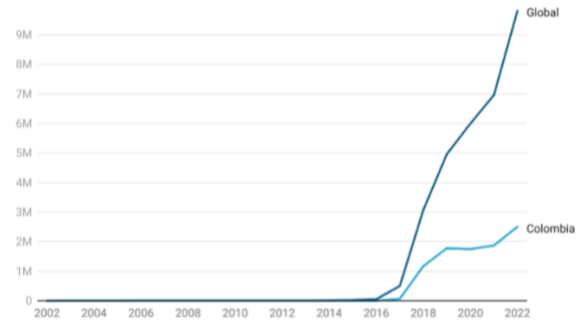


Figure 1. Migration of Venezuelans Over the Past 20 Years.

While it can seem like an exaggeration to refer to slang, like *veneco/a*, as discriminatory xenophobic rhetoric, it is appropriately classified as hate speech by meeting the following markers: (1) it provokes hatred and violence to large populations; (2) uses social media as medium; (3) is used broadly to describe groups; and (4) often used in rhetoric [12]. Scholars of linguistics, culture, and communication have started to take notice and offer an explanation for this shift. A sociologist at Universidad del Norte explains that when people are grouped with an adjective, even if it is a word derived from their nationality, it serves as a conduit of stigmatization and rejection. Now that there is a negative image about Venezuela, referring to Venezuelans even with a word like *veneco* derived as a shortened term for their nationality, will reproduce a negative association [10]. In this way, the word changes from a colloquialism without predetermined connotation to a pejorative.

Analyzing this ‘othering’ of Venezuelan migrants by Colombians serves to create a hierarchal power structure between two socio-cultural groups; othering by Colombians situates Venezuelan migrants as separate and below them in their societal order. This phenomenon of intergroup othering is not new or isolated to this particular case. Situating migrants as ‘invaders’ has been used to legitimize state-sanctioned violence in places like India, for decades where women who migrate from other regions are assumed to be involved in sex work, out of desperation for resources [13]. Joysheel Shrivastava [14] examines the prevalence of this ‘othering’ amongst women in India, where she describes women as the primary perpetrators in continuing a traditional violent language that is rooted in patriarchy.

Methodology

The proposed approach consists of seven major steps summarized in Figure 2. The first four steps encompass (1) the extraction of the tweets, (2) geolocation to identify whether a Tweet is from Colombia or not, (3) validation to only select the Tweets related to the Venezuelan migration and discard the ones about other topics, and (4) the identification of tone and topic. The following three steps contain the (5) temporal and (6) spatiotemporal analysis of the tone of the conversation, and (7) a temporal analysis of the pejorative terms *veneco* and *veneca*.

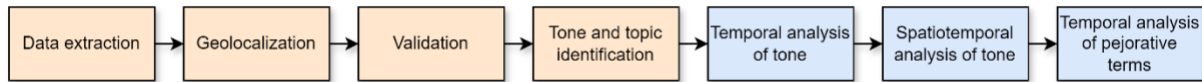


Figure 2. Methodology.

1. Data extraction

Using the Twitter Streaming API, we ran a search query containing keywords related to the Venezuelan migration accounting for their contracted versions and plurals. In addition, we included the variations by grammatical gender (feminine and masculine) since in Spanish the nouns have gender. We extracted the queries containing the keywords, but also the Tweets they referenced either as a retweet or a mention. In total, 7.4 million tweets were

extracted using the query in Figure 3, in the range of January 2014 and June 2022. We did not remove duplicates to keep the additional effect that one tweet can have when appearing multiple times.

veneco OR veneca OR venecos OR venecas OR ((venezolano OR venezolana OR venezolanos OR venezolanas OR venezuela OR vzla OR vnzla) AND (migrante OR migrantes OR migracion))

Figure 3. Tweets extraction query.

2. Geolocalization

From the extracted tweets, in 2.35 million, the geolocation was activated by the user, and in 5.1 million, it was deactivated. Such geolocation, however, was mentioned as a raw text with no discrimination between different geographic levels, such as countries or cities. For instance, in “Bogotá” it is not explicitly mentioned that it is within the country of Colombia. Using the Nominatim API [15], we automatically converted the plain text to a country location. From the 2.35 million, the most frequent countries were Venezuela (47%), Colombia (15.1%), the United States (6.4%), and Spain (4.5%). As we wanted to capture the whole conversation in Colombia, we decided to create a classification model whose purpose was to identify whether a tweet was from Colombia or not based on the tweet’s text. With this, even when a Tweet had no geolocation activated, we could identify if it was from Colombia. We fine-tuned three models: spaCy es_core_news_sm [16] and the transformer BETO [17], a BERT model trained on a sizable Spanish corpus (since all our tweets were in Spanish). Fine-tuning involves copying the weights from a pre-existing model and adjusting them for a new task [18]. To create the training and testing dataset, we randomly selected the text and country from 135,000 tweets from Colombia and another 135,000 from any other country, resulting in a total dataset of 270,000 tweets. Using a 70/30 split, we created the training (189,000) and testing (81,000) datasets. This ratio was used because it is commonly used for classification tasks with neural networks, [19] showing that it provides the best performance compared to other ratios in their task. The training results are presented in Table 1.

Table 1. Performance of the text geolocation models.

Architecture	Accuracy
es_core_news_sm	0.79
BETO	0.50

spaCy es_core_news_sm was selected to automatically label the entire dataset due to its highest accuracy score. After inference, from the 5.1 million Tweets with geolocation deactivated, 1.65 million were geolocated to Colombia based on its text.

3. Validity and tone identification

To ensure that our analysis focused exclusively on tweets relevant to the Venezuelan migration, we developed a classification model to differentiate between pertinent and irrelevant tweets. We began by extracting a subset of 4,684 tweets from our dataset for manual labeling. Each tweet was independently reviewed and categorized as either relevant (discussing Venezuelan migration) or irrelevant (unrelated topics). This binary classification resulted in 2,991 tweets labeled as relevant and 1,693 as irrelevant.

We addressed class imbalance and duplication issues to prepare the data for model training. We removed duplicate tweets from the relevant category and performed random equal balancing to create a balanced dataset. This process yielded 1,554 tweets in each category, totaling 3,108 tweets. We then divided this dataset into a training and testing sets using a 70/30 split. The training set comprised 2,170 tweets, while the testing set contained 938 tweets.

We fine-tuned three different models to classify tweet relevance: the two previously utilized models and the transformer BERT-base-multilingual-uncased-sentiment [20]. The performance accuracy metrics are presented in Table 2. The model spaCy es_core_news_sm was selected to label the entire dataset due to its highest accuracy score in both tasks (0.79 and 0.90, respectively).

Table 2. Performance of the Validation models.

Architecture	Accuracy
es_core_news_sm	0.90
BETO	0.81
BERT-base multilingual-uncased-sentiment	0.84

Pejorative terms’ tone identification

Understanding the sentiment and tone associated with the use of pejorative terms is crucial for analyzing xenophobic discourse. We focused on the terms “veneco” and “veneca,” which have evolved into derogatory slurs in the Colombian context. From the 2,991 tweets previously identified as relevant, we further analyzed the sentiment associated with these terms.

Two independent coders manually labeled the tweets to classify the subject matter and the tone of the terms “veneco” and “veneca.” The tone was assessed using a Likert scale ranging from -3 (extremely negative) to +3 (extremely positive), with 0 representing a neutral tone. The labeling criteria were established through team discussions and consensus, as illustrated in Figure 4.

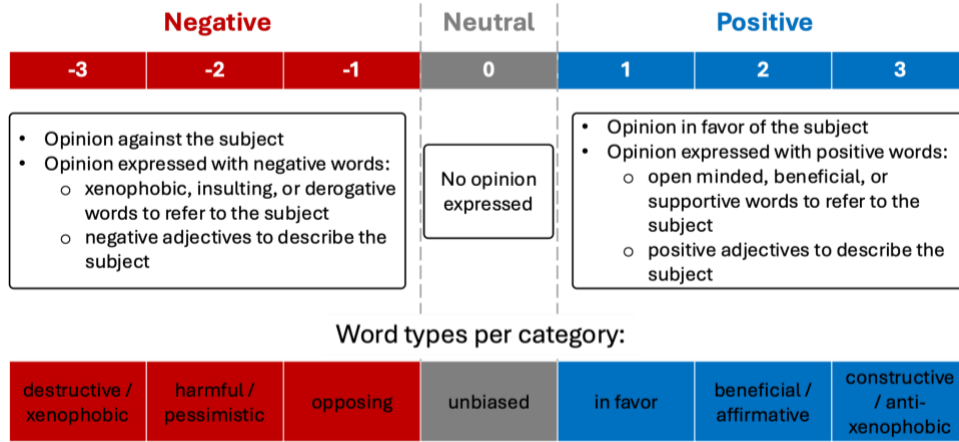


Figure 4. Criteria used for the tone scale labeling.

Intercoder reliability was evaluated, resulting in an agreement rate of 60%. While this level of agreement may seem moderate, it is considered acceptable in the context of sentiment analysis due to the subjective nature of interpreting emotional expressions in text [21]. Differences in individual perceptions can affect consistency, and a 60% agreement reflects the inherent challenges in manually labeling sentiment.

To maintain objectivity, one of the two labeled datasets was randomly selected for subsequent analysis. After filtering and preparing the data, we obtained a dataset of 1,920 tweets. We divided this dataset into training and testing sets using a 70/30 split, resulting in 1,344 tweets for training and 576 for testing, as summarized in Table 3.

Table 3. Performance of the tone classification models.

Model	Training size	Testing size	Total size
Tone	1344	576	1920

Results

1. Temporal analysis of tone

Figure 5 shows the proportion of Tweets with every type of tone where the average proportion was higher for negative tweets (44%), followed by neutral (35%) and positive (21%). From the total number of tweets, the

proportion of retweets is 61%, and Figure 6 shows the proportion only for the unique tweets, discarding the retweets. The mean proportion of negative tweets increased to 55%, neutrals decreased to 25% and positives remained in 20%. The impact of retweets in the tone proportion can be noticeable given the portion of negative tweets and was driven by an increase in the portion of neutral tweets. This suggests that users more often retweet positive or neutral content and that increase in the total number of Tweets counters the effect of negative Tweets. This is particularly true when we identify that 67% of retweets had a non-negative tone.

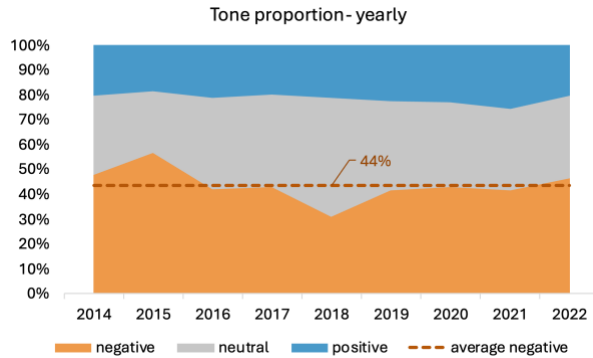


Figure 5. Tone yearly proportion.

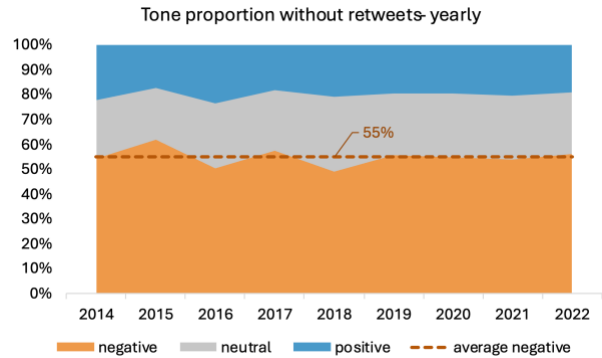


Figure 6. Tone yearly without retweets.

The frequency of Tweets with specific tones (Figure 7) shows us that while in mid-2015 and the beginning of 2017 experienced peaks in negative tone, the conversation was smaller in comparison with the period of 2018-2022. There, again the negative tweets were predominant in the majority of months with the exception of the period September 2018 (also the month with the highest number of tweets of the whole timeframe) to June 2019 where the negative was more frequent.

When looking at the conversation during individual months one can identify that the tone is mostly driven by two factors: the events that happen within the month, and the number of retweets related to such events. For instance, when analyzing the peaks for neutral, negative, and positive tones we identified: (1) in 2018-09, the month with the highest neutral tweets, 42% of the total tweets referred to the same event that on September 5, a woman and her mom were undressed by Venezuelan guards. While this event was labeled neutral by the model, considering the topic this would be actually negative. (2) In 2019-11, the month with the highest negative tweets, around 9% of the tweets were around the expulsion of Venezuelans and overall, this month contains a high number of tweets with xenophobic comments against the Venezuelans, relating them to crimes and the increase in insecurity and destruction. This was also caused by the event where according to Colombia's migration agency, 59 Venezuelans were expelled by boat on Monday after allegedly vandalizing the country, causing violence and disturbances during the marches with thousands of Colombians [22]. (3) In 2021-02, the month with the highest positive tweets, more than 10% of the tweets had a positive tone because the president of Colombia at that time took the decision of giving temporal protection to Venezuelans (ETPV). Based on the tweets, this decision was well-received by Colombians, Venezuelan migrants, and other organizations.

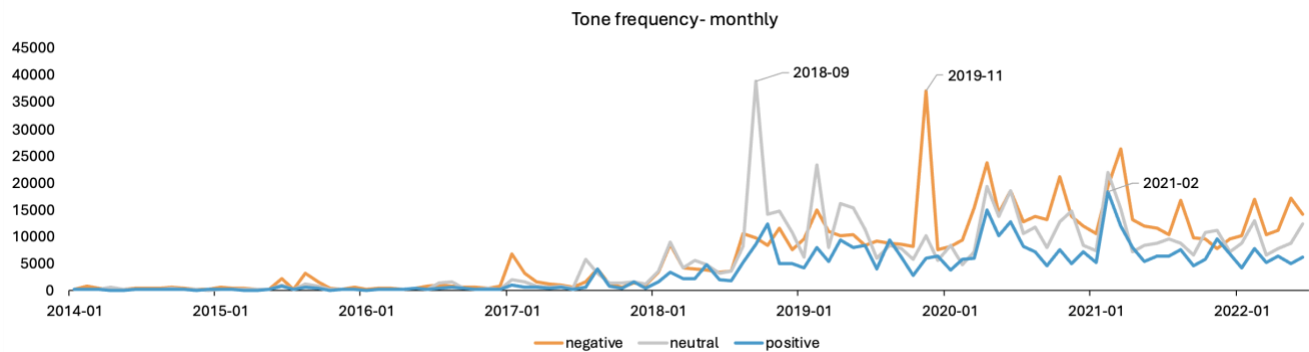


Figure 7. Tone monthly frequency.

However, when the retweets are discarded (Figure 8), the negative tone exhibits the highest frequency for all months and the negative and positive are noticeably similar. The number of unique tweets grows overall because for all tones there is a positive trend to increase, however, the speed to which the negative grows is considerably higher, more than two times, the growth of the neutral and positive. Showing that as time passed, more unique negative tweets were created. With this it is possible to conclude that the retweets serve a role of stabilizing the tone, showing a more realistic approximation of the population's point of view.

Additionally, in November 2019, October 2020, and March 2021 the highest negative tones were achieved. When contrasting their prevalence to when the retweets are included, only the first peak in November 2019 was retweeted by lots of people in the conversation.

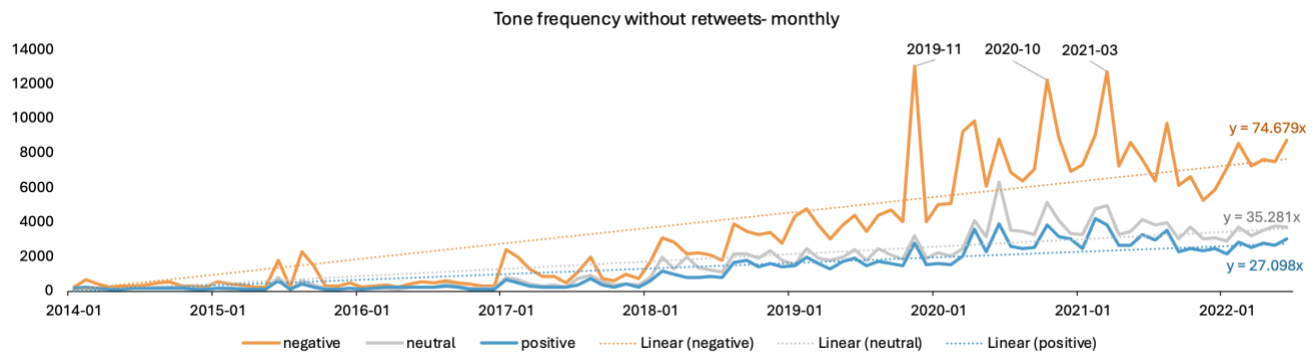


Figure 8. Tone monthly frequency without retweets.

2. Spatiotemporal analysis of tone

We explored whether the tone of tweets varied depending on their geographic origin within Colombia. Utilizing the latitude and longitude coordinates associated with each tweet, we determined the corresponding department from which the tweet was posted. This geospatial classification enabled us to analyze regional differences in sentiment towards Venezuelan migrants over time. To simplify our analysis, we focused on the six departments with the highest numbers of Venezuelan migrants between 2018 and 2022. Earlier years were excluded due to lower tweet frequencies, which could lead to less reliable conclusions.

The six departments analyzed were Bogotá, D.C., Valle del Cauca, Santander, Atlántico, Antioquia, and Norte de Santander. Figure 9 illustrates the monthly frequency of tweets classified by tone (positive, neutral, negative) for each department during the specified period. While all departments exhibited a similar overall trend of negative tweets being the most prevalent, notable variations were observed in the temporal patterns of sentiment across departments. For instance, the sentiment trends in Bogotá, D.C., closely mirrored the national average, indicating consistent public discourse in the capital region. In contrast, departments such as Valle del Cauca, Santander, and Atlántico displayed distinct patterns, including multiple peaks in negative sentiment during certain months.

These variations suggest that regional factors may influence public sentiment towards Venezuelan migrants, potentially due to local events, media coverage, or socio-economic conditions unique to each department. Understanding these differences is crucial for developing targeted interventions and policies that address xenophobia and promote social integration at the regional level.

When analyzing the impact of retweets on the tone distribution within each department, we observed effects consistent with our national-level findings. The inclusion of retweets tended to decrease the overall proportion of negative tweets while increasing the proportion of neutral tweets. This phenomenon can be attributed to the amplification of particular messages through retweeting, which may alter the perceived sentiment of the discourse. To quantify this effect, we calculated the difference in the proportion of tweets by tone for each department when retweets were included versus when they were excluded. Table 4 presents these differences, showing that, on average, the proportion of neutral tweets increased by 4.3%, correlating with a corresponding decrease in the proportion of negative tweets by -3.6%. This shift suggests that retweeted content is more likely to be neutral or less negative, potentially moderating the overall sentiment expressed within each department.

For example, in Bogotá D.C., including retweets resulted in a 7.0% increase in neutral tweets and a 7.0% decrease in negative tweets, with no change in positive tweets. Similar patterns were observed in other departments, albeit with varying magnitudes. This consistent trend across multiple regions underscores the role of retweeting behavior in shaping the sentiment landscape on Twitter.

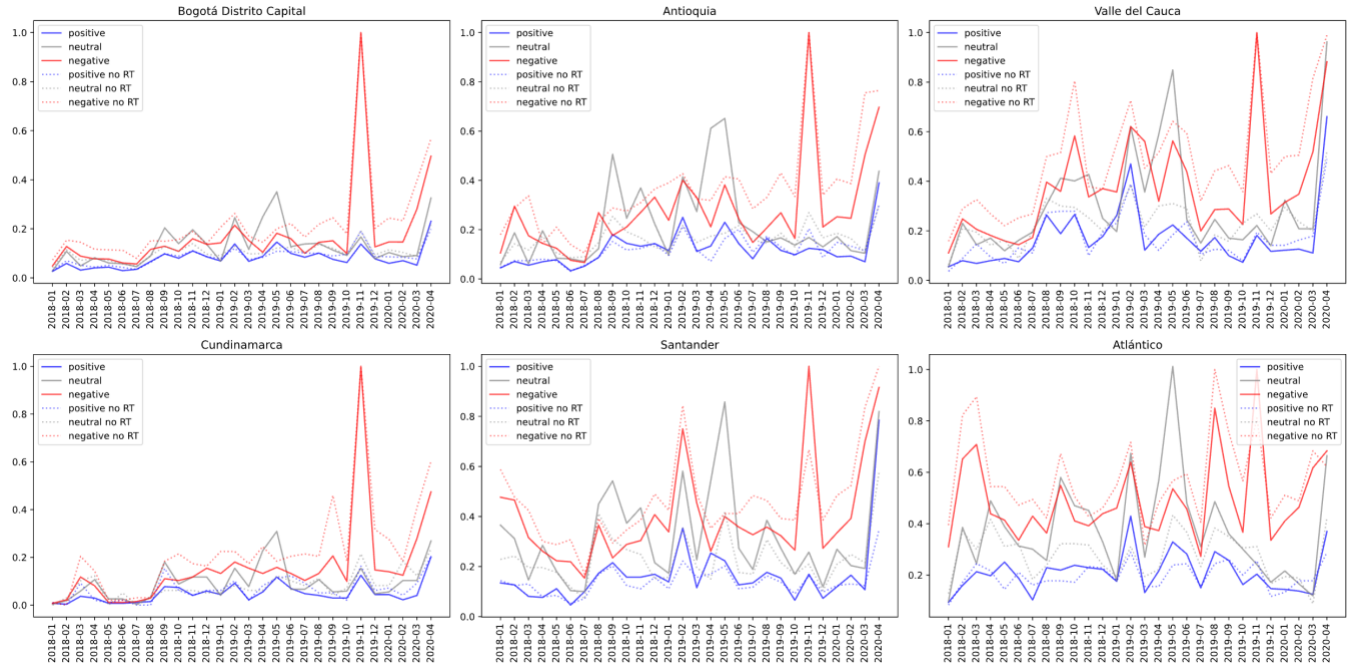


Figure 9. Tone frequency over time for the six departments with the most Venezuelan migrants.

Table 4. Proportion of tone difference per department when including Retweets.

Department	Negative	Neutral	Positive
Bogotá Distrito Capital	-7.0%	7.0%	0.0%
Bolívar	-5.5%	6.7%	-1.2%
Norte de Santander	-4.1%	5.0%	-0.9%
Boyacá	-9.8%	12.3%	-2.5%
Cesar	-6.6%	6.0%	0.6%
Antioquia	-6.9%	7.0%	-0.1%
Santander	-6.7%	5.8%	0.8%
Quindío	-3.0%	3.2%	-0.2%
Valle del Cauca	-6.4%	6.0%	0.5%
Atlántico	-5.2%	5.4%	-0.2%
Risaralda	-2.3%	3.0%	-0.6%
Huila	-5.2%	7.1%	-1.9%
Magdalena	-5.9%	8.3%	-2.4%
Córdoba	-2.8%	3.9%	-1.1%
Nariño	-5.2%	4.9%	0.3%
Tolima	-1.8%	2.7%	-0.9%
Cundinamarca	-2.0%	4.0%	-2.0%
La Guajira	-1.5%	3.1%	-1.6%
Meta	-3.2%	4.6%	-1.4%
Casanare	-4.5%	4.3%	0.3%
Arauca	-3.7%	5.2%	-1.5%

Chocó	-0.4%	0.7%	-0.3%
Caldas	-4.0%	4.3%	-0.3%
Vichada	-1.4%	2.1%	-0.8%
Guaviare	-0.4%	-0.4%	0.8%
Cauca	-3.3%	5.3%	-1.9%
Caquetá	0.0%	1.6%	-1.6%
Sucre	-2.3%	2.6%	-0.3%
Amazonas	-3.1%	4.8%	-1.7%
Guainía	-1.6%	2.2%	-0.6%
Putumayo	-2.6%	3.3%	-0.7%
Vaupés	0.0%	0.0%	0.0%
San Andrés y Providencia	0.0%	0.0%	0.0%

3. Temporal Analysis of Pejorative Terms

a. Terms usage estimation

The monthly and yearly distribution of tweets of the terms *veneco* and *veneca* are presented in Figure 10.

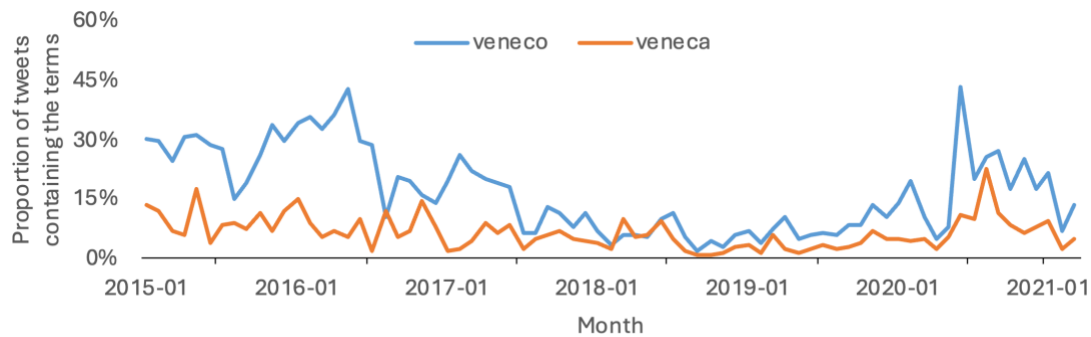


Figure 10. Monthly proportion of tweets containing the terms *veneco* and *veneca*.

For the majority of periods, the frequency of tweets containing *veneco* is higher than those containing *veneca*. This could be a result of masculine plurals including mixed male/female groups and is thus more encompassing of a term. Additionally, the proportion of tweets that contain either term is relatively higher in some years than others. Specifically, in 2015, the proportion was at its maximum for *veneco* (25%) and *veneca* (8%), followed by a decrease until 2018 to 5% and 2.5%, respectively. This also corresponds to a time when we observe an increase in the number of Venezuelans arriving in Colombia (see Figure 10), but the rate of arrival tapered off in later years. In 2021, the proportions rose again, possibly related to strained resources globally arising from COVID-19 and pandemic shutdowns. These strained resources and social tensions during COVID-19 could result in the scapegoating of migrants, thereby leading to increased hate speech towards Venezuelan migrants at that time.

If we relate the proportion of the term's usage to the number of Venezuelan migrants in Colombia, it is rather unusual that the proportion during the period from 2015 to 2018 is significantly high. Given that a substantial increase in the number of Venezuelan migrants does not occur until 2018.

The monthly proportion of negative tone for both terms is presented in. In the given context, the tones refer to the emotional tone or sentiment expressed in the tweets containing the terms. The tones can be informative as they provide insight into how the public perceives the terms and the events related to them. For instance, a high proportion of negative tones in tweets containing these terms could suggest negative attitudes towards Venezuelans or the humanitarian crisis.

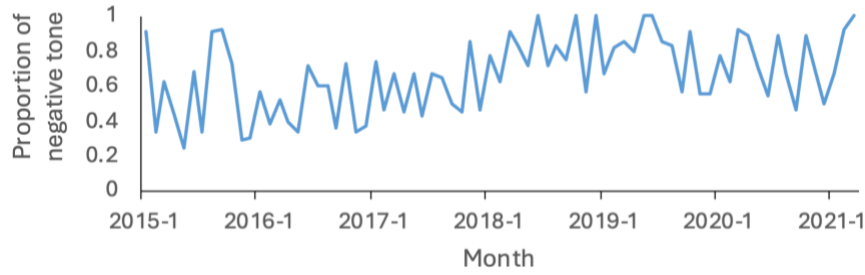


Figure 11. Monthly proportion of negative tone of the term *veneco*.

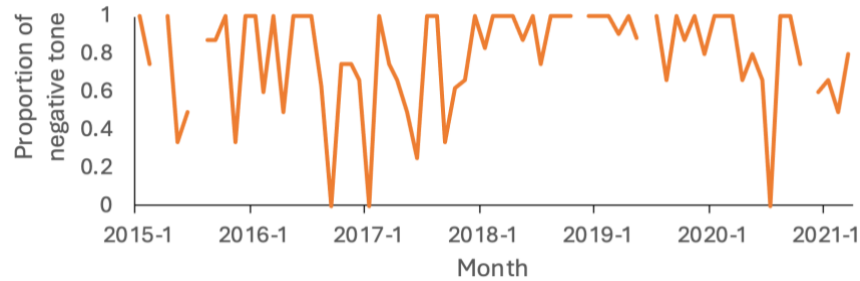


Figure 12. Monthly proportion of negative tone of the term *veneca*.

The two terms exhibit a predominantly negative tone, and it is noteworthy that the average negative tone associated with *veneca* is higher compared to that of *veneco*. Specifically, *veneca* has a 79% negative tone, whereas *veneco* reflects a 66% negative tone. In interviews we conducted with subject matter experts (linguists, sociologists, and communications scholars) who studied derogatory language in Colombia in 2023, one possible explanation for this is that the term *veneca* has always had a negative connotation, well beyond the recent humanitarian migration of Venezuelans. We describe this phenomenon in more detail below, as the related themes are reflected in our dataset. The series of instances involving *veneco* initially displayed a balanced distribution between negative and non-negative sentiments until 2018, coinciding with the significant increase in the number of Venezuelan migrants (see Figure 1). This observation shows that despite the highest usage of *veneco* and *veneca* occurring between 2015 and 2018, the associated predominant tone was not necessarily negative. This finding also aligns with statements in the literature, which suggest that the term *veneco* was originally used as a descriptor but has evolved over time to become a pejorative term.

On the other hand, the series of instances involving *veneca* exhibits high variability due to the limited number of observations. This variability may affect the reliability of the negativity measurement

Conclusions

Acknowledgments

The Large Language Model o1-preview from OpenAI was used to review the coherence and grammar of this manuscript.

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